

1. Administrative & Study Identification

Full Study Title: A PHASE 1/2 RANDOMIZED, OBSERVER-BLIND STUDY TO EVALUATE THE SAFETY, TOLERABILITY, AND IMMUNOGENICITY OF A MODIFIED RNA VACCINE AGAINST VARICELLA ZOSTER VIRUS IN HEALTHY INDIVIDUALS **Study Title / Short Title:** A Study to Learn About a Modified RNA Vaccine Against Shingles in Healthy Adults **Protocol Number:** C5031001 **NCT Number:** NCT05703607 **PostingID:** 386171666667166764 **Sponsor Name:** Pfizer (Collaborator: BioNTech) **Company:** Pfizer **Investigational Product:** VZV modRNA vaccine (BNT167) **Phase of Development:** Phase 1/Phase 2 **Phase ID:** PHASE2 **Phase Requirement (Criteria Type):** PHASE_REQUIREMENT **Criteria Text:** Trial must be in PHASE2 **Trial Status:** RECRUITING **Status Name:** RECRUITING **Status Description:** Currently recruiting participants **Medical Monitor:** Pfizer Clinical Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and immunogenicity of a modified RNA (modRNA) vaccine candidate against the varicella zoster virus (VZV) in healthy adults. Phase 1 focuses on selecting the optimal mRNA vaccine candidate, dose level, dosing schedule, and formulation for advancement to Phase 2. **Secondary Objectives:** To determine the persistence of immunity (duration of protection) and assess long-term safety over an extended follow-up period of up to 5 years.

2.2 Background & Rationale Shingles (herpes zoster or HZ) is a debilitating infectious disease caused by the reactivation of the varicella zoster virus (VZV). The condition manifests as an extremely painful localized rash and can lead to persistent complications such as postherpetic neuralgia (PHN). While approved vaccines for shingles exist, their administration is commonly associated with high reactogenicity. Pfizer and BioNTech aim to utilize their proprietary mRNA technology to develop a new shingles vaccine that provides high efficacy and long-lasting immunity while being efficiently produced and better tolerated by patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (saline) and Dose-selection controlled **Blinding:** Observer-Blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Targeted Enrollment: 900 evaluable patients **Number of Sites:** Multicenter (Conducted in the United States) **Estimated Study Start:** February 2023 **Estimated Primary Completion:** Ongoing

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy adults aged 50 to 85 years at the time of screening. 2. Has a Body Mass Index (BMI) between 18 and 35 kg/m² (inclusive) at the screening visit.

4.2 Exclusion Criteria 1. History of Herpes Zoster (shingles) or previous vaccination with any varicella or HZ vaccine. 2. Immunocompromised individuals with known or suspected immunodeficiency, or known infection with HIV, HCV, or HBV. 3. Prior history of heart disease (e.g., heart failure, recent coronary artery disease, cardiomyopathies, pericarditis, or myocarditis) or clinically relevant abnormalities on screening 12-lead ECG.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: The vaccine is administered either as a 2-dose series (dosed 2-months apart or 6-months apart) or as a single active VZV modRNA vaccine dose at the first visit followed by a saline placebo at the second vaccination visit. **Route of Administration:** Intramuscular (IM) injection. **Duration of Treatment:** 8 to 12 months for the initial assignment group, with selected groups continuing into a persistence-of-immunity portion lasting up to 5 years.

5.2 Concomitant & Prohibited Therapy Permitted: Participation in observational studies. **Prohibited:** Immunosuppressive therapy (e.g., cytotoxic agents or systemic corticosteroids); receipt of blood/plasma products or immunoglobulin within 60 days before intervention; other RNA vaccines within 28 days prior; participation in strenuous or endurance exercise within 7 days before or after each study administration.

5.3 Associated Drug Entities *Note: The following reference drug entity is mapped to the trial records:* **Drug Name:** phenoxymethylpenicillin **Drug Preferred Name:** phenoxymethylpenicillin **ATC Code:** J01CE02 **Trade Names:** Crystapen V, Truxcillin-VK, Suspen, V-Cil-K, Pen-V, Tenkicin, A-Cillin, Beepen-VK, Ledercillin VK, PC Pen VK, Pen-Vee K, Veetids, V-Cillin K **Source Level:** 5

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Blood/Chemistry Labs, 12-lead ECG, Troponin Assessment
Baseline	Day 0	Randomization, First Dose (Vaccination 1), E-diary setup
Interim	Month 2 or 6	Second Dose (Vaccination 2 or Saline Placebo), Safety & Immunogenicity Labs
End of Study	Month 8-12 to Year 5	Persistence of Immunity assessment (Cellular/Humoral), Long-term Safety Monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- SSA:** Percentage of participants reporting local reactions: Pain at the injection site, redness, and swelling as self-reported in electronic diaries.
- SSA:** Percentage of participants reporting systemic events: Fever, fatigue, headache, chills, vomiting, diarrhea, new or worsened muscle pain, and new or worsened joint pain, as self-reported in electronic diaries.
- SSA:** Percentage of participants reporting adverse events: As elicited by investigational site staff.

7.2 Secondary & Exploratory Outcomes

- SSA:** Geometric mean concentrations (GMCs) of glycoprotein E antibodies in proportion of evaluable immunogenicity participant.
- SSA:** Geometric mean fold rise (GMFR) from before vaccination to each subsequent timepoint in evaluable immunogenicity participants.
- SSA:** Proportion of evaluable immunogenicity participants with vaccine response in glycoprotein E antibodies from baseline to each subsequent timepoint.

* Additional goals include optimal mRNA candidate, dose level, and formulation selection (Phase 1), and persistence of immunity over 5 years.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Daily collection via patient electronic diaries (7-day post-vaccination reactogenicity), physical examinations, and continuous monitoring for immune-mediated conditions.

Serious Adverse Events (SAEs): Standard reporting timeline within 24 hours of site awareness.

Data Monitoring Committee (DMC): Internal/External safety review committee for dose-escalation decisions and Phase 2 candidate selection.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary immunogenicity outcomes; Safety Set for all reactogenicity and

AE/SAE evaluations. **Sample Size Justification:** Target of $N=900$ is powered to provide robust safety, tolerability, and immunogenicity data across multiple candidate formulations and dose schedules to inform Phase 2 selection. **Handling of Missing Data:** Mixed models for repeated measures (MMRM) for longitudinal immunogenicity endpoints; missing safety diary data is generally not imputed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring to assure protocol compliance, cardiac safety checks (ECG and troponin tracking), and proper cold-chain management of the mRNA vaccine.

Audit Trail: Secure eCRF locking, tracking of biological samples, and verified data clarification forms.

11. Study Identifiers & Governance

Primary ID: C5031001 **Registry Number:** NCT05703607 **Sponsor/Lead**

Organization: Pfizer (in partnership with BioNTech) **Study Title:**

A Study to Learn About a Modified RNA Vaccine Against Shingles in Healthy Adults **Official Regulatory Title:** A PHASE 1/2 RANDOMIZED, OBSERVER-BLIND STUDY TO EVALUATE THE SAFETY, TOLERABILITY, AND IMMUNOGENICITY OF A MODIFIED RNA VACCINE AGAINST VARICELLA ZOSTER VIRUS IN HEALTHY INDIVIDUALS

12. Clinical Framework (Disease/Vaccine Specifics)

Primary Condition / Name: Herpes Zoster Infection **Disease Areas:**

Shingles, Herpes Zoster Infection, Human **Preferred UMLS Name:**

Herpes Zoster Infection **Preferred Name:** Herpes zoster (disorder)

Semantic Type: Disease or Syndrome **Definition:** A common dermal and neurologic disorder caused by reactivation of the varicella-zoster virus that has remained dormant within dorsal root ganglia, often for decades, after the patient's initial exposure to the virus in the form of varicella (chickenpox). It is characterized by severe neuralgic pain along the distribution of the affected nerve and crops of clustered vesicles over the area.

MeSH Code: D006562 **SNOMED ID:** 4740000 **Medical Methodology:**

Prophylactic mRNA vaccination targeting VZV glycoprotein E (gE)

Optimization Target: High efficacy with lower reactogenicity compared to existing protein-adjuvant vaccines

13. Study Procedures & Methodology

Management Pathway: Screening of healthy volunteers, administration of interventional vaccine, and longitudinal tracking of immune response **Education Protocol (HCP):** Proper handling and administration of lyophilized/modRNA formulations, cardiac safety screening **Education Protocol (Patient):** Patient training on accurate reactogenicity reporting via electronic diaries **Quality Audit Mechanism:** Active cardiac surveillance (ECG) and strict review of immune-mediated conditions prior to and after vaccination

14. Observation & Follow-Up Schedule

Total Duration: Up to 5 years for persistence-of-immunity cohort **Onsite Visit Cadence:** Up to 13 visits over the primary 8-month period; possibility of a 5-year extension with an additional 5 visits **Remote Monitoring Frequency:** Intermittent follow-up phone calls or electronic reviews to capture emergent health changes **Checklist Review Interval:** E-diary review immediately following each vaccination dose

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				